

56. First number = $(50 \times 2) = 100$.

$$\text{Second number} = \left(\frac{50 \times 250}{100} \right) = 125.$$

57. L.C.M. = $\frac{\text{Product of numbers}}{\text{H.C.F.}} = \frac{1320}{6} = 220$.

58. H.C.F. of co-prime numbers is 1. So, L.C.M. = $\frac{117}{1} = 117$.

59. Since H.C.F. is always a factor of L.C.M., we cannot have three numbers with H.C.F. 35 and L.C.M. 120.

60. H.C.F. of two numbers divides their L.C.M. exactly. Clearly, 8 is not a factor of 60.

61. The factors of 9570 are 2, 3, 5, 11 and 29. Clearly, H.C.F. can be any of these factors occurring not more than once and no number other than 2, 3, 5, 11 or 29 or having any factor other than these, can be the H.C.F. So, the only possibility is 11.

62. Clearly, the numbers are (23×13) and (23×14) .

$$\therefore \text{Larger number} = (23 \times 14) = 322.$$

63. Since 16 is not a factor of 136, it follows that there does not exist any pair of numbers with H.C.F. 16 and L.C.M. 136.

64. Product of numbers = $21 \times 4641 = 97461$.

Let the numbers be $21a$ and $21b$.

$$\text{Then, } 21a \times 21b = 97461 \Rightarrow ab = 221.$$

Now, co-primes with product 221 are (1, 221) and (13, 17).

So, the numbers are $(21 \times 1, 21 \times 221)$ and $(21 \times 13, 21 \times 17)$.

Since one number lies between 200 and 300, the suitable pair is (273, 357).

65. Product of numbers = 29×4147 .

Let the numbers be $29a$ and $29b$.

$$\text{Then, } 29a \times 29b = (29 \times 4147) \Rightarrow ab = 143.$$

Now, co-primes with product 143 are (1, 143) and (11, 13).

So, the numbers are $(29 \times 1, 29 \times 143)$ and $(29 \times 11, 29 \times 13)$.

Since both numbers are greater than 29, the suitable pair is $(29 \times 11, 29 \times 13)$, i.e., (319, 377).

$$\therefore \text{Required sum} = (319 + 377) = 696.$$

66. H.C.F. of two prime numbers is 1.

$$\text{Product of numbers} = (1 \times 161) = 161.$$

Let the numbers be a and b . Then, $ab = 161$.

Now, co-primes with product 161 are (1, 161) and (7, 23).

Since x and y are prime numbers and $x > y$, we have $x = 23$ and $y = 7$.

$$\therefore 3y - x = (3 \times 7) - 23 = -2.$$

67. H.C.F. of 2436 and 1001 is 7. Also, H.C.F. of 105 and 7 is 7.

$$\therefore \text{H.C.F. of } 105, 1001 \text{ and } 2436 \text{ is } 7.$$

68. For the minimum number of rows, the number of trees in each row must be the maximum.

$$\therefore \text{Number of trees in each row} = \text{H.C.F. of } 21, 42, 56 = 7.$$

Hence, number of rows

$$= \left(\frac{21 + 42 + 56}{7} \right) = \frac{119}{7} = 17.$$

69. Required length = H.C.F. of 700 cm, 385 cm and 1295 cm = 35 cm.

70. Required capacity

$$= \text{H.C.F. of } 120 \text{ litres and } 56 \text{ litres}$$

$$= 8 \text{ litres} = 8000 \text{ cc.} \quad [\because 1 \text{ litre} = 1000 \text{ cc}]$$

71. Maximum possible daily wage = H.C.F. of ₹ 5750 and ₹ 5000 = ₹ 250.

72. For the least number of bottles, the capacity of each bottle must be maximum.

$$\therefore \text{Capacity of each bottle} = \text{H.C.F. of } 403 \text{ litres, } 465 \text{ litres and } 496 \text{ litres} = 31 \text{ litres.}$$

Hence, required number of bottles

$$= \left(\frac{403 + 465 + 496}{31} \right) = \frac{1364}{31} = 44.$$

73. Required number of students = H.C.F. of 1001 and 910 = 91.

74. Largest size of the tile = H.C.F. of 378 cm and 525 cm = 21 cm.

75. For the least number of tiles, the size of the tile must be the maximum.

$$\text{Maximum size of the tile} = \text{H.C.F. of } 1517 \text{ cm and } 902 \text{ cm} = 41 \text{ cm.}$$

Hence, required number of tiles

$$= \frac{\text{Area of ceiling}}{\text{Area of each tile}} = \left(\frac{1517 \times 902}{41 \times 41} \right) = 814.$$

76. Number of books in each stack = H.C.F. of 336, 240 and 96 = 48.

$$\text{Hence, total number of stacks} = \left(\frac{336 + 240 + 96}{48} \right) = \frac{672}{48} = 14.$$

77. Maximum length of each part = H.C.F. of 78 cm, 104 cm, 117 cm, 169 cm = 13 cm.

$$\therefore \text{Number of pieces} =$$

$$\left(\frac{78 + 104 + 117 + 169}{13} \right) = \frac{468}{13} = 36.$$

78. Required number

$$= \text{H.C.F. of } (91 - 43), (183 - 91) \text{ and } (183 - 43)$$

$$= \text{H.C.F. of } 48, 92 \text{ and } 140 = 4.$$

79. $d = \text{H.C.F. of } (8506 - 7654),$

$$(9997 - 8506) \text{ and } (9997 - 7654)$$

$$= \text{H.C.F. of } 852, 1491 \text{ and } 2343 = 213.$$

Clearly, $r = 199$.

$$\therefore d - r = 213 - 199 = 14.$$

80. $N = \text{H.C.F. of } (4665 - 1305), (6905 - 4665) \text{ and } (6905 - 1305)$

$$= \text{H.C.F. of } 3360, 2240 \text{ and } 5600 = 1120.$$

$$\text{Sum of digits in } N = (1 + 1 + 2 + 0) = 4.$$

81. Required volume

$$= [\text{H.C.F. of } (129 - 57), (177 - 129)$$

$$\text{and } (177 - 57)] \text{ litres}$$

$$= (\text{H.C.F. of } 72, 48 \text{ and } 120) \text{ litres} = 24 \text{ litres.}$$

2	9570
3	4785
5	1595
11	319
	29

213	7654	35
	639	
	1264	
	1065	
	199	